Foundations of Programming Languages

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1 Abstract syntax trees

An *abstract syntax tree* (AST) is a tree representation of the hierarchical structure of a program. Unlike a concrete parse tree, an AST omits purely syntactic detail such as parenthesis, semicolons used as separators, keywords whose meaning is already captured by the node type; retaining only the information needed to define the program's meaning.

Each *internal node* of the AST corresponds to a language construct (sequence, assignment, conditional, etc.) and each *leaf* corresponds to an atomic value or reference.

For an imperative language with mutable references, the relevant constructs are:

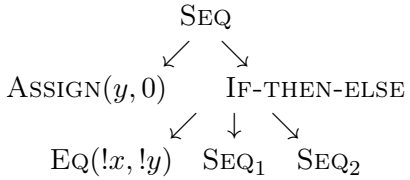
- $C_1 \,;\, C_2$, sequential composition
- $l := e$, assignment to location l
- **if** B **then** C_1 **else** C_2, conditional command
- **while** B **do** C, iteration
- $!l$, dereferencing a location (read the value stored at l)
- n, b, integer and boolean literals
- e_1 *op* e_2, arithmetic or boolean combination

1.1 solved example

Build the AST of the following program:

$y := 0 \,;$ **if** $!x = !y$ **then** $x := !x + 1 \,;\; y := !y - 1$ **else** $x := !x - 1 \,;\; y := !y + 1$

The top level operator is the semicolon that separates the two main components, so the root is a SEQ node. Its left child is the assignment $y := 0$, and its right child is the conditional. The conditional has three children: the guard $!x = !y$, the *then*-branch $x := !x + 1\,;\; y := !y - 1$, and the *else*-branch (symmetric).



note that $!x$ is represented as a DEREF node applied to the location x, which is distinct from x as an identifier: the location names a cell in the store, and dereferencing reads its content.

1.2 Exercises

- Build the AST of: $x := !x \times 2\;;$ **if** $!x > 10$ **then skip else** $x := !x + 1$
- Build the AST of: **while** $!i < !n$ **do** $s := !s + !i\,;\; i := !i + 1$
- Reconstruct the source program corresponding to the following AST: root is SEQ with left child ASSIGN$(x, 5)$ and right child IF-THEN-ELSE(EQ$(!x, 5)$, ASSIGN$(y, 1)$, Ass

2 Type inference

Type inference is the process by which a compiler deduces the type of an expression without explicit annotations from the programmer. the classical algorithm for the simply typed lambda calculus (and its extensions) is *Hindley Milner* (HM), which works by:

1. Assigning a fresh type variable α_i to each sub-expression whose type is unknown.
2. Generating *constraints* of the form $\tau_1 = \tau_2$ from the typing rules.
3. Solving the constraints by *unification*.

if unification fails, the program contains a type error; if it succeeds, the resulting substitution gives the most general type.

2.1 Typing rules for references

$$\text{T-Deref}\ \frac{\Gamma \vdash !l : T}{\Gamma \vdash l : \textbf{ref}\ T} \qquad \text{T-Assign}\ \frac{\Gamma \vdash l := e : \textbf{unit}}{\Gamma \vdash l : \textbf{ref}\ T \quad \Gamma \vdash e : T} \qquad \text{T-While}\ \frac{\Gamma \vdash \textbf{while}\ B\ \textbf{do}\ C : \textbf{unit}}{\Gamma \vdash B : \textbf{bool} \quad \Gamma \vdash C : \textbf{unit}}$$

2.2 solved example

$\lambda f.\, f\ f$ does not typecheck

Consider `proc(f)zero?((f f))`. We try to infer the type of f.

- Let $f : \alpha$.
- The expression $f\ f$ requires f to be a function type, so $\alpha = \alpha \to \beta$ for some fresh β.
- But $\alpha = \alpha \to \beta$ has no finite solution: expanding repeatedly yields $\alpha = (\alpha \to \beta) \to \beta = ((\alpha \to \beta) \to \beta) \to \beta = \cdots$

unification fails (occurs check). Therefore the expression has no type in the simply typed lambda calculus without recursive types.

2.3 Scala generics

```
case class MyPair[A,B](x:A,y:B)
val p = MyPair(1, "Scala")
def id[1](x:T)=x
val q = id(1)
```

the definition of `id` introduces a `polymorphic` type: it can be applied to any type , and each call site independently instantiates a T.

```
B  = !n < 5
B1 = !term <= 2
C1 = res := !n
C2 = c:=!a+!b; res:=!n; !c; a:=!b; b:=!c
C  = if B1 then C1 else C2; n:=!n+1
a:=1; b:=1; while B do C
```

reading off types from the operations performed:

- n, res, a, b, c : **ref int** (assigned integers and dereferenced in arithmetic).
- *term* : **ref int** (compared with 2).

- $B, B_1 : \mathbf{bool}$.
- $C_1, C_2, C : \mathbf{unit}$ (commands produce no value).
- The whole program : **unit**.

2.4 Exercises

Infer the type of the S combinator: $\lambda x.\, \lambda y.\, \lambda z.\; x\, z\, (y\, z)$. What constraints does unification generate?

Consider $\lambda f.\, \lambda x.\; f\, (f\, x)$. Does it typecheck? If so, give its most general type.

What types are inferred for `f`, `result`, and `pairs` in the following Scala snippet?

```
def f[A](lst: List[A]) = lst.head
val result = f(List(1,2,3))
val pairs  = List((1,"a"), (2,"b"))
```

Given $x : \mathbf{ref\ int}$ and $b : \mathbf{ref\ bool}$, does the following expression typecheck? If so, give its type.

$$b :=\!!x = 0 \;;\; \mathbf{if}\ !b\ \mathbf{then}\ x := 1\ \mathbf{else\ skip}$$

3 Operational semantics

3.1 Big-step semantics

In *big-step* (natural semantics, a judgement) $\langle e,\, \sigma\rangle \Downarrow v$ asserts that expression e in store σ evaluates *directly* to value v, without exposing intermediate steps. Derivations are trees whose leaves are axioms and whose internal nodes are inference rules.

3.1.1 Lists: `concat` and `append`

We assume the constructors **nil** (empty list) and $\mathbf{cons}(h, t)$ (list with head h and tail t).

`concat`(e, ℓ), insert element e at the end of ℓ.

Concat-Nil
$$\langle \mathbf{concat}(e, \mathbf{nil}),\, \sigma\rangle \Downarrow \mathbf{cons}(e, \mathbf{nil})$$

Concat-Cons
$$\frac{\langle \mathbf{concat}(e, \mathbf{cons}(h, t)),\, \sigma\rangle \Downarrow \mathbf{cons}(h, t')}{\langle \mathbf{concat}(e, t),\, \sigma\rangle \Downarrow t'}$$

append(ℓ_1, ℓ_2) concatenate two lists.

$$\textsc{Append-Nil}\quad \langle \mathbf{append}(\mathbf{nil}, \ell_2),\, \sigma\rangle \Downarrow \ell_2 \qquad \textsc{Append-Cons}\quad \frac{\langle \mathbf{append}(\mathbf{cons}(h, t_1), \ell_2),\, \sigma\rangle \Downarrow \mathbf{cons}(h, \ell')}{\langle \mathbf{append}(t_1, \ell_2),\, \sigma\rangle \Downarrow \ell'}$$

Derivation for append We verify $\langle \mathbf{append}(\mathbf{cons}(1, \mathbf{nil}), \mathbf{cons}(2, \mathbf{nil})),\, \sigma\rangle \Downarrow \mathbf{cons}(1, \mathbf{cons}(2, \mathbf{nil}))$:

$$\dfrac{\textsc{Append-Cons}\quad \langle \mathbf{append}(\mathbf{cons}(1, \mathbf{nil}), \mathbf{cons}(2, \mathbf{nil})),\, \sigma\rangle \Downarrow \mathbf{cons}(1, \mathbf{cons}(2, \mathbf{nil}))}{\textsc{Append-Nil}\quad \langle \mathbf{append}(\mathbf{nil}, \mathbf{cons}(2, \mathbf{nil})),\, \sigma\rangle \Downarrow \mathbf{cons}(2, \mathbf{nil})}$$

3.2 Small-step semantics

In *small-step* semantics, a judgement $\langle P, \sigma\rangle \to \langle P', \sigma'\rangle$ performs a single atomic reduction. A program is fully evaluated when no rule applies (it is a value or a stuck state) or when it reduces to **skip** (for commands).

3.2.1 For-loops in SIMP

The grammar of SIMP extended with **for** is:

$$C ::= l := E \mid C\,;\, C \mid \mathbf{if}\ B\ \mathbf{then}\ C\ \mathbf{else}\ C \mid \mathbf{while}\ B\ \mathbf{do}\ C \mid \mathbf{skip} \mid \mathbf{for}\ (C;\ B;\ C)\ \{C\}$$

The cleanest approach is to treat **for** as *syntactic sugar*:

$$\mathbf{for}\ (C_i;\ B;\ C_s)\ \{C_b\} \triangleq C_i\,;\, \mathbf{while}\ B\ \mathbf{do}\ C_b\,;\, C_s$$

If the language requires explicit small-step rules (no sugar), we give:

$$\textsc{For-Init}\quad \frac{\langle \mathbf{for}\ (C_i;\ B;\ C_s)\ \{C_b\},\, \sigma\rangle \to \langle \mathbf{for}\ (C_i';\ B;\ C_s)\ \{C_b\},\, \sigma'\rangle}{\langle C_i,\, \sigma\rangle \to \langle C_i',\, \sigma'\rangle}$$

$$\textsc{For-Skip}\quad \langle \mathbf{for}\ (\mathbf{skip};\ B;\ C_s)\ \{C_b\},\, \sigma\rangle \to \langle \mathbf{if}\ B\ \mathbf{then}\ (C_b\,;\, C_s\,;\, \mathbf{for}\ (\mathbf{skip};\ B;\ C_s)\ \{C_b\})\ \mathbf{else}\ \mathbf{skip},\, \sigma\rangle$$

Rule For-Init reduces the initialiser one step at a time. Rule For-Skip fires once the initialiser has been fully executed: it checks the guard and either runs the body followed by the step expression (and loops), or terminates with **skip**.

3.3 Exercises

- Write big-step rules for $\mathbf{reverse}(\ell)$, which reverses a list.
- Use `concat` as a helper.

  Prove with a full derivation tree:

  $$\langle \mathbf{concat}(3, \mathbf{cons}(1, \mathbf{cons}(2, \mathbf{nil}))), \sigma \rangle \Downarrow \mathbf{cons}(1, \mathbf{cons}(2, \mathbf{cons}(3, \mathbf{nil})))$$

- Write big-step rules for $\mathbf{length}(\ell)$, which returns the number of elements, and evaluate $\langle \mathbf{length}(\mathbf{cons}(1, \mathbf{cons}(2, \mathbf{nil}))), \sigma \rangle \Downarrow 2$ with a derivation tree.

4 Lambda calculus

Lambda calculus is a formal system that serves as the theoretical foundation for functional programming languages. The core idea is surprisingly simple: everything reduces to functions, how they are defined and how they are applied.

4.1 basic syntax

There a re only three constructs in the language:

- **Variables:** $x, y, z \ldots$
- **Abstraction:** $\lambda x.\, e$ — defines a function with parameter x and body e.
- **Application:** $e_1\, e_2$ — applies function e_1 to argument e_2.

from just these three pieces, any computation can be expressed.

4.2 Free and Bound Variables

In $\lambda x.\, M$, we say x is **bound**: the abstraction defines its scope. A variable that appears outside the scope of any λ is called **free**.

α-equivalence tells us that the name of a bound variable does not matter, as long as renaming causes no conflicts:

$$\lambda x.\, x \;\equiv_\alpha\; \lambda y.\, y$$

4.3 Reduction Rules

Computation happens through two main rules:

- β**-reduction** — substitutes the argument into the function body:

$$(\lambda x.\, e)\, v \;\to\; e[x := v]$$

  This is the fundamental computational step.

- η**-reduction** — two functions are equal if they produce the same result for every argument:

$$\lambda x.\, (f\, x) \;\to\; f \qquad (\text{if } x \text{ is not free in } f)$$

4.4 Evaluation Strategies

In what order do we reduce? There are several options, each with different consequences:

1. **Normal Order:** always reduces the leftmost, outermost redex first. Guaranteed to find a normal form if one exists.
2. **Call-by-Value:** evaluates the argument before passing it to the function (used in C, Java, Python).
3. **Call-by-Name:** passes the argument unevaluated — the basis for lazy evaluation.

4.5 Church Encodings

Since lambda calculus has no built-in types or literals, we need to *encode* familiar values as functions:

- **Booleans:**

$$\begin{aligned} TRUE &\equiv \lambda t.\, \lambda f.\, t \\ FALSE &\equiv \lambda t.\, \lambda f.\, f \\ AND &\equiv \lambda p.\, \lambda q.\, p\, q\, p \end{aligned}$$

- **Church Numerals** — the number n applies f exactly n times:

$$\begin{aligned} 0 &\equiv \lambda f.\, \lambda x.\, x \\ 1 &\equiv \lambda f.\, \lambda x.\, f\, x \\ 2 &\equiv \lambda f.\, \lambda x.\, f\, (f\, x) \\ SUCC &\equiv \lambda n.\, \lambda f.\, \lambda x.\, f\, (n\, f\, x) \end{aligned}$$

4.6 Recursion: The Y Combinator

Since functions are anonymous, recursion is achieved via fixed points. Curry's Y Combinator is defined as:

$$Y = \lambda f.(\lambda x.f(x\,x))(\lambda x.f(x\,x))$$

It satisfies the property: $Y\,g = g(Y\,g)$, enabling infinite recursion.

4.7 Recursion: The Y Combinator

Lambda functions are anonymous, so recursion is not straightforward. The solution is Curry's **fixed-point combinator**:

$$Y = \lambda f.\,(\lambda x.\,f\,(x\,x))\,(\lambda x.\,f\,(x\,x))$$

It satisfies $Y\,g = g\,(Y\,g)$, which allows recursive functions to be defined without ever naming them.

5 Type safety

Type safety is the formal guarantee that well-typed programs do not (go wrong), they never reach a state that is undefined or meaningless. It is established through two theorems that work together: **progress** and **preservation**.

5.1 progress

A well-typed expression is never *stuck*, it is either already a value, or it can take a step.

Teorema 5.1 (progress)**.** *If* $\vdash e : \tau$ *and* $e \to e'$*, then* $\vdash e' : \tau$*.*

Intuitively, this rules out situations like applying an integer as if it were a function, a well-typed program always has somewhere to go.

5.2 preservation

Taking a step does not change the type of an expression.

Teorema 5.2 (preservation)**.** *If* $\vdash e : \tau$ *and* $e \to e'$*, then* $\vdash e' : \tau$

Together, progress and preservation give us the **type safety** property: evaluation of a well-typed expression either produces a value of the expected type, or diverges; but it never crashes.

5.3 Supporting lemmas

The proofs of the both theorems rely on two key lemmas.

Lema 5.3 (Canonicity)**.** *If* $\vdash v : \tau$ *and* v *is a value, then the* shape *of* v *is determined by* τ*. For example:*

- *If* $\tau = Bool$*, then* v *is either true or false.*
- *If* $\tau = \tau_1 \to \tau_2$*, then* $v = \lambda x.\, e$ *for some* x *and* e*.*

This lemma is what lets us case-split on the shape of a value during a proof, we know exactly what forms are possible.

Lema 5.4 (Substitution)**.** *If* $\Gamma, x : \tau' \vdash e : \tau$ *and* $\vdash v : \tau'$*, then* $\Gamma \vdash e[x := v] : \tau$*.*

in plain terms, substituing a well-typed value for a variable preserves typing. This is the key step in proving Preservation for β reduction.

5.4 proof strategy: structural induction

Both theorems are proved by **structural induction** on the typing derivation. The general scheme is:

1. **Base cases:** handle values and variables directly, these are straightforward since no reduction is possible.
2. **Inductive cases:** for each typing rule (application, abstraction, conditionals, etc.), assume the result holds for the subexpressions (*induction hypothesis*), then show it holds for the whole expression.
3. **Use the lemmas:** Canonicity tells you what shape a value has; Substitution handles the β-reduction case in Preservation.

the structure of the proof closely mirrors the structure of the language, one case per typing rule. Once you see the patternm it becomes mechanical.

6 Denotational semantics

While operational smeantics describes *how* a program executes step by step, denotational semantics asks a different question: *what does a program mean?*. the idea is to assign each program a mathematical object, its denotation, in a way that is compositional; the meaning of compound expression is built entirely from the meanings of its parts.

6.1 interpretation functions

The central tool is the **interpretation function** $[\![\cdot]\!]$, which maps syntactic expressions to mathematical values. The double brackets are just notation, is like a translation from syntax into math.

6.2 the state domain

Programs do not compute in a vacuum, they read and write variables. We model this with a *state*, a snapshot of memory at any point during execution:

$$\Sigma = \mathit{Var} \to \mathit{Val}$$

A state $\sigma \in \Sigma$ assigns a value to every variable. With this, we can give meaning to both expressions and commands.

6.3 meaning of expressions

An arithmetic or boolean expression takes a state and returns a value:

$$[\![e]\!] : \Sigma \to \mathit{Val}$$

For example, the meaning of $x + 1$ in state σ is just $\sigma(x) + 1$. There is no mutation here, expressions are pure functions over states.

6.4 meaning fo commands

Commands are more interesting: they *transform* states. A command takes an initial state and produces a final one:

$$[\![C]\!] : \Sigma \to \Sigma_\perp$$

The $\perp$ subscript is important. It accounts for *non-termination*, a `while` loop may never finish, so the result is not always a state. $\Sigma_\perp = \Sigma \cup \{\perp\}$ adds a special element $\perp$ ("bottom") to represent divergence.

6.5 fixed points and iteration

The `while` loop is the tricky case. We cannot define its meaning directly, since it may run for an unbounded number of steps. instead, we use **Kleene's fixed-point theorem**.

The idea is to build the meaning of `while` b `do` C as the least fixed point of a functional Φ:

$$\Phi(f)(\sigma) = \begin{cases} f([\![C]\!](\sigma)) & \text{if } [\![b]\!](\sigma) = \mathit{true} \\ \sigma & \text{if } [\![b]\!](\sigma) = \mathit{false} \end{cases}$$

We approximate the meaning from below: $\Phi^0 = \bot$, $\Phi^1 = \Phi(\bot)$, $\Phi^2 = \Phi(\Phi(\bot))$, and so on. The least fixed point $\mathit{lfp}(\Phi) = \bigsqcup_{n \geq 0} \Phi^n$ captures *all finite unrollings* of the loop, which is exactly what the loop computes.

6.6 equivalence with operational semantics

A natural question arises: do both styles of semantics agree? The answer is yes.

Teorema 6.1 (Equivalence)**.** *For all commands C, initial states σ, and final states σ':*

$$\langle C, \sigma \rangle \Downarrow \sigma' \iff [\![C]\!](\sigma) = \sigma'$$

operational semantics tells us *how* the program runs, and denotational semantics tells us *what* it computes, and they agree on every input/output pair. Neither one is more "correct"; they are two lenses on the same truth.

7 Axiomatic semantics

Axiomatic semantics takes a different approach from the other two styles: instead of describing how a program executes or what function it computes, it gives meaning to programs through **logical assertions** about their behavior. The central idea is to reason about what is true *before* and *after* a program runs.

7.1 Hoare triples

The basic unit is the **Hoare triple**:

$$\{P\}\ C\ \{Q\}$$

where P is the **precondition**, C is a command, and Q is the **postcondition**. Read it as: "if P holds before running C, and C terminates, then Q holds afterward."

For example:

$$\{x = 5\}\ x := x + 1\ \{x = 6\}$$

7.2 inference rules

The power of hoare logic comes from its **compositional rules**, one per language construct.

- **Assignment:**

$$\{P[x := e]\}\ x := e\ \{P\}$$

  Work backwards: to have P after the assignment, you need P with e substituted for x before it.

- **Sequence:**

$$\frac{\{P\}\ C_1\ \{R\} \quad \{R\}\ C_2\ \{Q\}}{\{P\}\ C_1;\ C_2\ \{Q\}}$$

  The postcondition of the first command becomes the precondition of the second.

- **Conditional:**

$$\frac{\{P \wedge b\}\ C_1\ \{Q\} \quad \{P \wedge \neg b\}\ C_2\ \{Q\}}{\{P\}\ \texttt{if}\ b\ \texttt{then}\ C_1\ \texttt{else}\ C_2\ \{Q\}}$$

  Each branch gets the precondition strengthened with the condition or its negation.

- **While:**

$$\frac{\{I \wedge b\}\ C\ \{I\}}{\{I\}\ \texttt{while}\ b\ \texttt{do}\ C\ \{I \wedge \neg b\}}$$

  The loop **invariant** I must hold before the loop, be preserved by every iteration, and together with the negated condition gives the postcondition when the loop exits.

- **Consequence:**

$$\frac{P \Rightarrow P' \quad \{P'\}\ C\ \{Q'\} \quad Q' \Rightarrow Q}{\{P\}\ C\ \{Q\}}$$

  Allows strengthening preconditions and weakening postconditions, essential for connecting rules together in a proof.

7.3 loop invariants

Finding a good loop invariant is often the hardest part od a hoare logic proof. A valid invariant I must satisfy three conditions:

1. I holds **before** the loop begins.
2. If I holds and the condition b is true, then after one iteration of the body, I still holds.
3. When the loop exits (b is false), $I \wedge \neg b$ implies the desired postcondition.

7.4 partial vs. total correctness

There are two flavors of correctness:

- **Partial correctness** $\{P\}\, C\, \{Q\}$: if C terminates, then Q holds. It says nothing about *whether* C terminates.
- **Total correctness** $[P]\, C\, [Q]$: C is guaranteed to terminate *and* Q holds afterward.

total correctness requieres an additional argument, typically a **variant**, a quantity that strictly decreases with each iteration and is bounded below.

7.5 weakness preconditions

Rather tahn guessing a precondition, we can *compute* the weakest one that makes a triple valid. The **weakest precondition** $wp(C, Q)$ is the least restrictive P such that $\{P\}\, C\, \{Q\}$ holds.

The key cases are:

$$wp(x := e,\ Q) = Q[x := e]$$
$$wp(C_1;\ C_2,\ Q) = wp(C_1,\ wp(C_2, Q))$$
$$wp(\texttt{if}\ b\ \texttt{then}\ C_1\ \texttt{else}\ C_2,\ Q) = (b \Rightarrow wp(C_1, Q)) \wedge (\neg b \Rightarrow wp(C_2, Q))$$

For `while` loops, wp cannot be computed mechanically in general, this is where loop invariants come back in.

7.6 Examples

7.6.1 Factorial

Consider a program that computes $n!$ iteratively, storing the result in r. A suitable invariant is:

$$I \ : \ r \cdot i! = n! \ \wedge \ 0 \leq i \leq n$$

Before the loop: $r = 1$, $i = n$, so $1 \cdot n! = n!$ — the invariant holds. Each iteration multiplies r by i and decrements i, preserving $r \cdot i! = n!$. When $i = 0$, we get $r = n!$.

7.6.2 Array Sum

For a program that sums the elements of an array $a[0..n-1]$ into variable s:

$$I \ : \ s = \sum_{k=0}^{i-1} a[k] \ \wedge \ 0 \leq i \leq n$$

The invariant tracks a partial sum. When the loop exits with $i = n$, we have $s = \sum_{k=0}^{n-1} a[k]$, which is exactly the full sum.